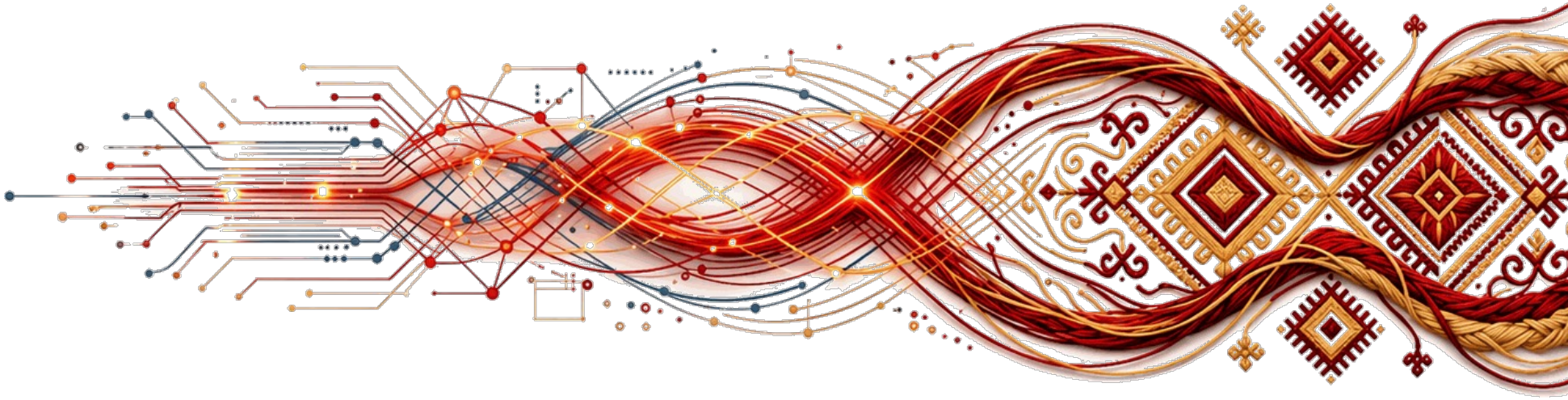
Introduction to Data Science

BASIC TRACK · LECTURE 3



Exploratory analysis, statistics and inference

Describing, visualising and testing data

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Vezilka
EuroHPC Joint Undertaking
Co-funded by the European Union
Republic of North Macedonia
Ministry of Digital Transformation

Agenda

1. **Describing one variable**
   Variable types, histograms, mean, median, spread, box plots
2. **Probability and the normal distribution**
   Probability as area, the 68–95–99.7 rule, z-scores
3. **Relationships and honest charts**
   Scatter plots, correlation, Simpson's paradox, misleading charts
4. **From a sample to the population**
   Sampling distributions, the CLT, the bootstrap, confidence intervals
5. **Hypothesis testing**
   The logic of a test, the p-value, two kinds of error, t and χ^2 tests
6. **AI-assisted exploratory analysis**
   Profiling and a review checklist

Learning objectives of the lecture

- After this lecture, the student is able to:
  - Name the **type of a variable** and read a **histogram** and a **box plot**.
  - Compute and interpret the **mean, median, standard deviation and quartiles**, and choose between them.
  - Use the **normal distribution** and z-scores, and explain the **central limit theorem** in words.
  - Interpret a **correlation coefficient**, recognise **Simpson's paradox** and correct a misleading chart.
  - Construct a **bootstrap confidence interval** and state correctly what it means.
  - Carry out the **t-test and the χ^2 test**, interpret the p-value, and explain type I and type II errors.
  - Review an AI-generated exploratory analysis with a checklist.

01

Describing one variable

Before any formula: what kind of variable is it, what does its distribution look like, and which numbers summarise it?



Key words and notation used in this lecture

Term	Meaning	Example: food deliveries
Observation	One measured case: a row of the data table	One delivery
Variable	A property measured on every observation: a column	Delivery time in minutes
Population	All units we want to know about	All deliveries in the city this year
Sample	The units we actually observed; their number is **n**	12 deliveries we timed, n = 12
Parameter	A number describing the population; Greek letters **μ, σ**	The true mean delivery time μ
Statistic	A number computed from the sample; Latin letters **$\bar{x}$, s**	The sample mean $\bar{x}$ = 22.8 min
Distribution	Which values occur and how often	Most take 15–25 min, a few take 40+
Σx_i	Add up all values: $x_1 + x_2 + \ldots + x_n$	12 + 14 + … + 48 = 274

Types of variables: categorical and numerical

Categorical: the values are labels

Numerical: the values are numbers

1 Nominal

Categories without an order.
Example: payment method (card, cash, voucher).
Summary: counts and %.

2 Ordinal

Categories with an order.
Example: customer rating, 1 to 5 stars.
Summary: counts, % and the median.

3 Discrete

Counts: whole numbers.
Example: number of items in an order.
Summary: mean or median, SD.

4 Continuous

Any value on a scale.
Example: delivery time in minutes.
Summary: mean or median, SD, quartiles.

- The type decides which arithmetic makes sense: if card = 1, cash = 2 and voucher = 3, the "average payment method" 1.8 is meaningless, although software will happily compute it.

Reading a distribution: from raw values to a histogram

- A *distribution* tells us **which values occur and how often**.
- A *histogram* splits the number line into equal **bins** and draws one bar per bin; the height of the bar is the **count** of values in that bin.
- The 12 delivery times in 5-minute bins: the bin 15–20 min holds **4** deliveries, and one delivery took 48 min.
- The bin width changes the picture: bins that are too wide hide detail, bins that are too narrow show noise.
- Always read the **axis labels and units** first.

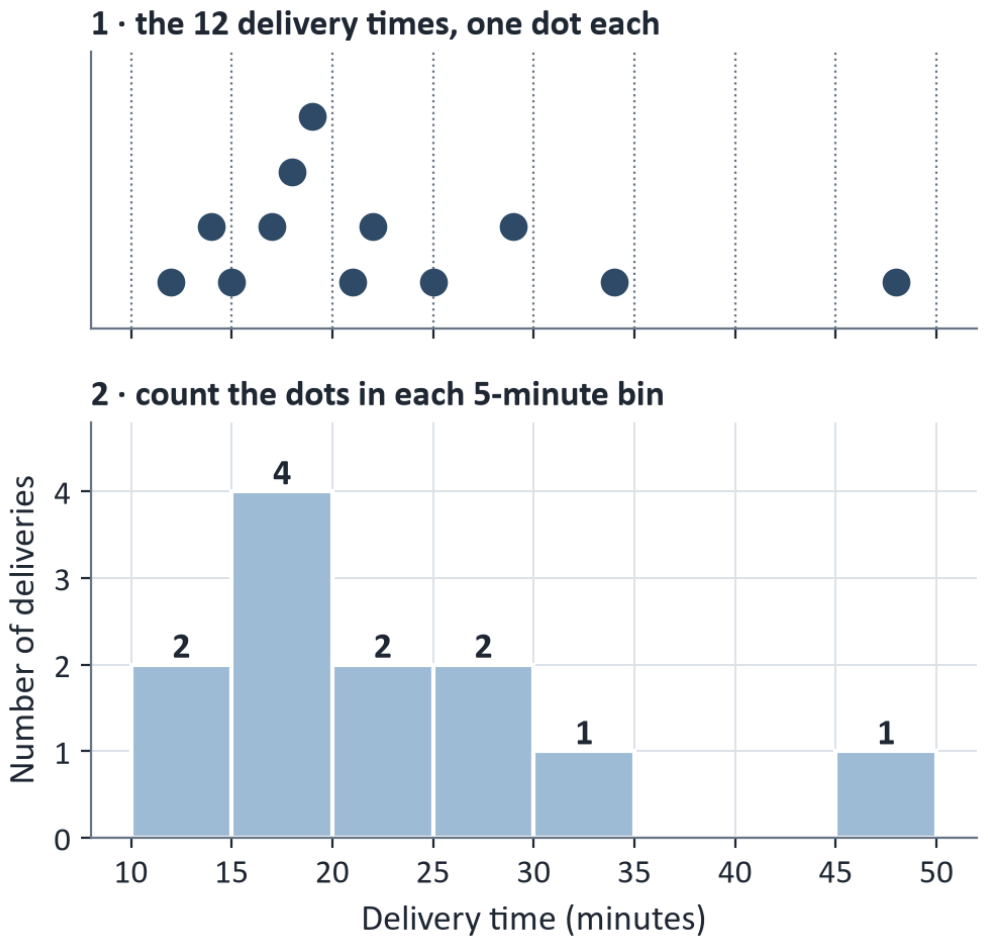


Figure 1 — building a histogram

The shape of a distribution: symmetry, skew and outliers

- *Symmetric*: the two halves are mirror images, and the mean ≈ the median.
- *Right-skewed*: a long tail of **large values** pulls the mean above the median: incomes, house prices, delivery times.
- *Left-skewed*: a long tail of **small values**: scores on an easy exam.
- An *outlier* is a value far from the rest; first check whether it is an **error or a real case**.
- The *skewness* g_1 summarises the asymmetry: about 0 when symmetric, positive for a right tail, negative for a left tail.

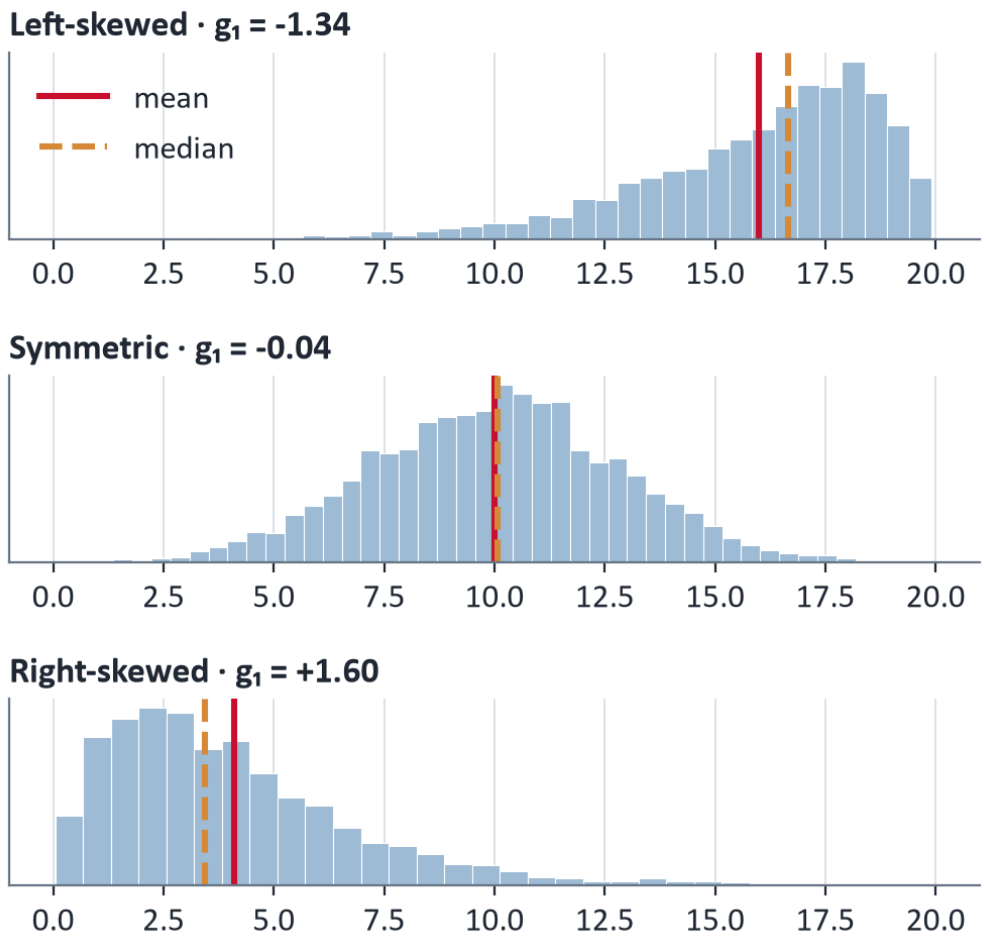


Figure 2 — red: mean; gold: median

Measures of location: the sample mean and the median

- The *mean* $\bar{x}$ adds all values and divides by their number: $\bar{x} = (x_1 + \ldots + x_n) / n$.
- The *median* is the **middle value** after sorting; when n is even, it is the average of the two middle values.
- Ages 17, 19, 21, 22, 23, 23, 23, 38: mean **23.25**, median (22 + 23) / 2 = **22.5**.
- Replace 38 by 68: the mean jumps to **27.00**, the median stays at **22.5**: the median *resists outliers*.
  - Salaries: nine employees earn 700–900 EUR and the owner 20,000 EUR; mean **2,720 EUR**, median **810 EUR**.
- The *mode* is the most frequent value (here 23); for nominal data it is the only possible centre.

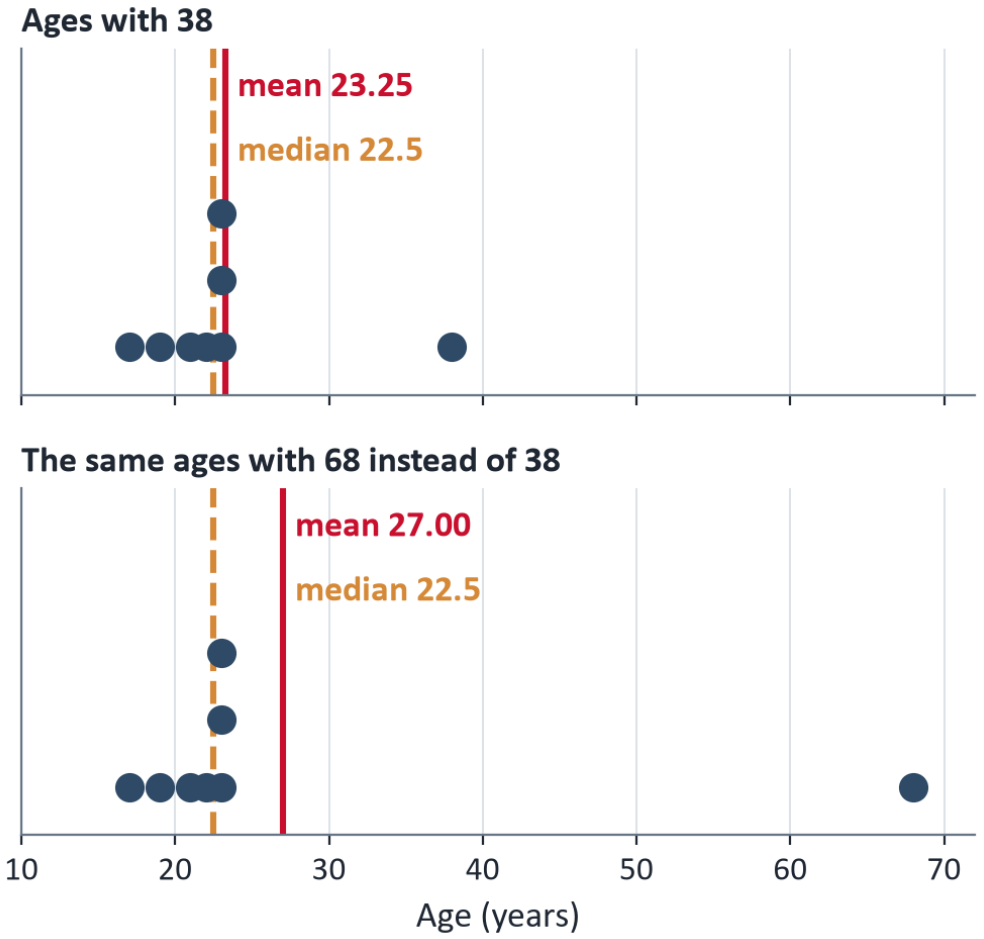


Figure 3 — mean sensitivity in this sample

Measures of spread: range, variance and standard deviation

- Two groups can have the **same mean** and still be very different: in the figure both have mean 50, with σ = 5 and σ = 15.
- The *range* = maximum − minimum: simple, but it depends on **only two values**.
- The *variance* s^2 is the average **squared distance from the mean** (dividing by n − 1).
- The *standard deviation* $s = \sqrt{s^2}$ is in the **units of the data**: a typical value lies about s away from the mean.
- Eight ages: range **21 years**, s = **6.34 years** (next slide).

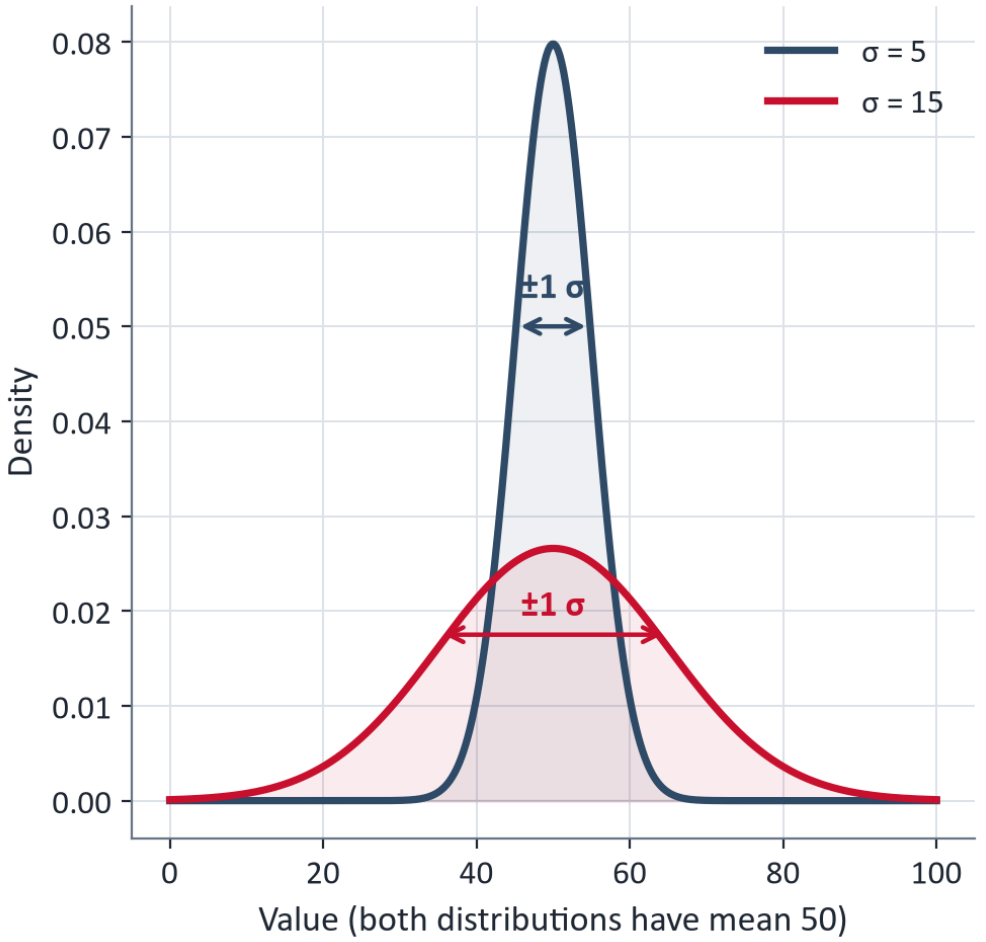


Figure 4 — equal means, different spreads

Worked example: the standard deviation of eight ages

Age x	$x - \bar{x}$	$(x - \bar{x})^2$
17	-6.25	39.06
19	-4.25	18.06
21	-2.25	5.06
22	-1.25	1.56
23	-0.25	0.06
23	-0.25	0.06
23	-0.25	0.06
38	+14.75	217.56
Σ = 186	0.00	**281.50**

1. **Mean**
   $\bar{x} = 186 / 8 = 23.25$
2. **Deviations**
   subtract $\bar{x}$ from every value; they always sum to 0
3. **Square and add**
   $\Sigma (x - \bar{x})^2 = 281.5$
4. **Divide by n − 1 = 7**
   $s^2 = 281.5 / 7 = 40.21$ years^2
5. **Square root**
   $s = \sqrt{40.21} = 6.34$ years

- The single value 38 contributes **217.56** of the 281.50: squaring makes large deviations dominate, which is why the SD, like the mean, is sensitive to outliers.

Quartiles, the box plot and percentiles

- Sort the data and cut it into **four equal parts**: the cut points are the *quartiles* Q_1, Q_2 (= the median) and Q_3.
- 12 delivery times: Q_1 = **16.5**, median **20**, Q_3 = **26** min; the *interquartile range* IQR = $Q_3 - Q_1$ = **9.5** min is the spread of the **middle half**.
- A *box plot*: a box from Q_1 to Q_3, a line at the median, whiskers to the last values within 1.5 · IQR; points beyond are flagged.
  - 48 min lies beyond Q_3 + 1.5 · IQR = 40.25 and is flagged; **flagged does not mean wrong**.
- "90 % of orders arrive within 30 min" is a statement about the **90th percentile (P90)**, not about the mean.

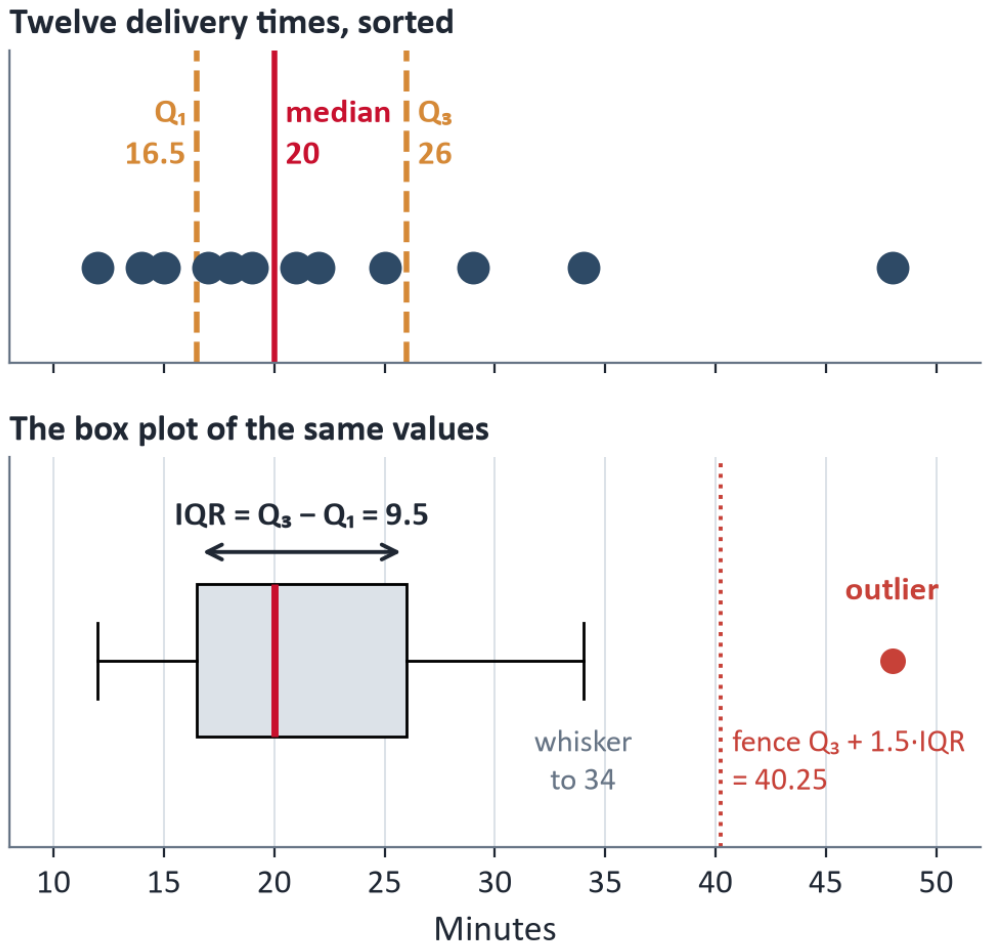
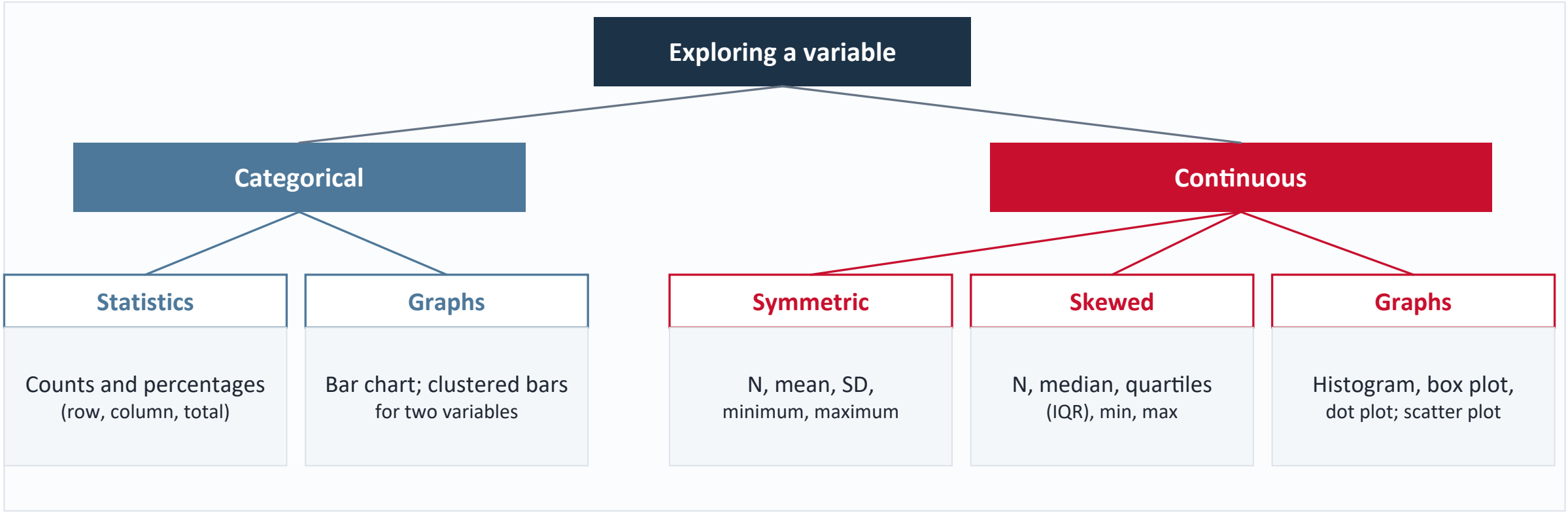


Figure 5 — quartiles and Tukey's box plot

Choosing descriptive statistics and graphs by variable type



- The type and the shape decide the summary: counts and % for categories, the mean and SD for symmetric numbers, the median and IQR for skewed numbers. When in doubt, report both and **show the plot**.

02

Probability and the normal distribution

Probability describes how often values occur in the long run; the normal curve is the most used model and the basis of the inference later in the lecture.



Probability: bars for counts, areas for measurements

- A *probability* is a number between 0 and 1: the share of times an outcome occurs **in the long run**.
- *Discrete* variable (items per order): each value has a probability, and the bars **sum to 1**.
- *Continuous* variable (height, time): probability is the **area under a curve**, the *density*.
  - P(170 ≤ height ≤ 180) = the shaded area ≈ **0.52**; the whole area is 1.
  - The height of the curve is **not** a probability; one exact value has probability 0.
- The *cumulative* probability F(x) = P(X ≤ x) is the area to the **left** of x.

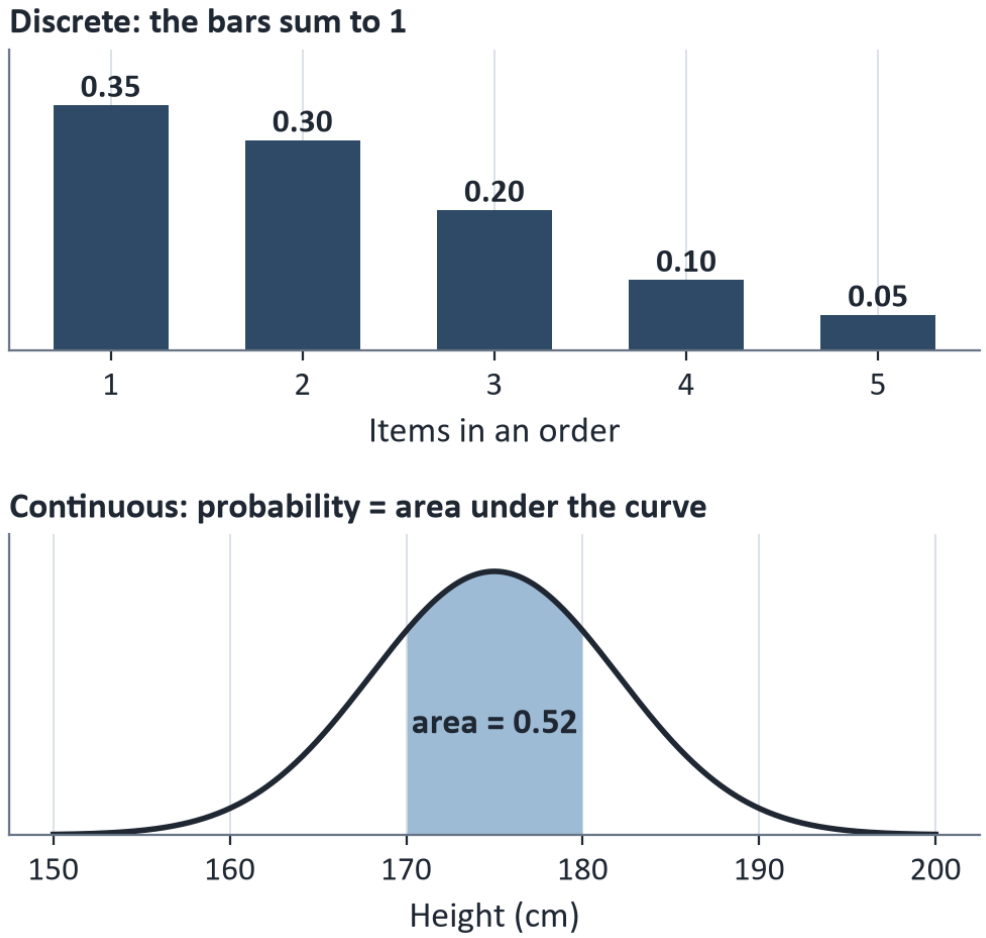


Figure 6 — probability as bar height and as area

The normal distribution and the 68–95–99.7 rule

- The *normal distribution* $N(\mu, \sigma^2)$ is the symmetric bell curve: μ sets the centre, σ the width.
- The **68–95–99.7 rule**: about 68 % of values lie within 1σ of μ, 95 % within 2σ, 99.7 % within 3σ.
  - Adult heights $\approx N(175, 7^2)$ cm: 68 % between **168 and 182**, 95 % between **161 and 189** cm.
- A *z-score* $z = (x - \mu) / \sigma$ counts **how many standard deviations** a value lies from the mean.
  - 189 cm → $z = (189 - 175) / 7 = \mathbf{2}$: taller than about 97.5 % of people.
- The rule holds for normal data only: check the histogram first.

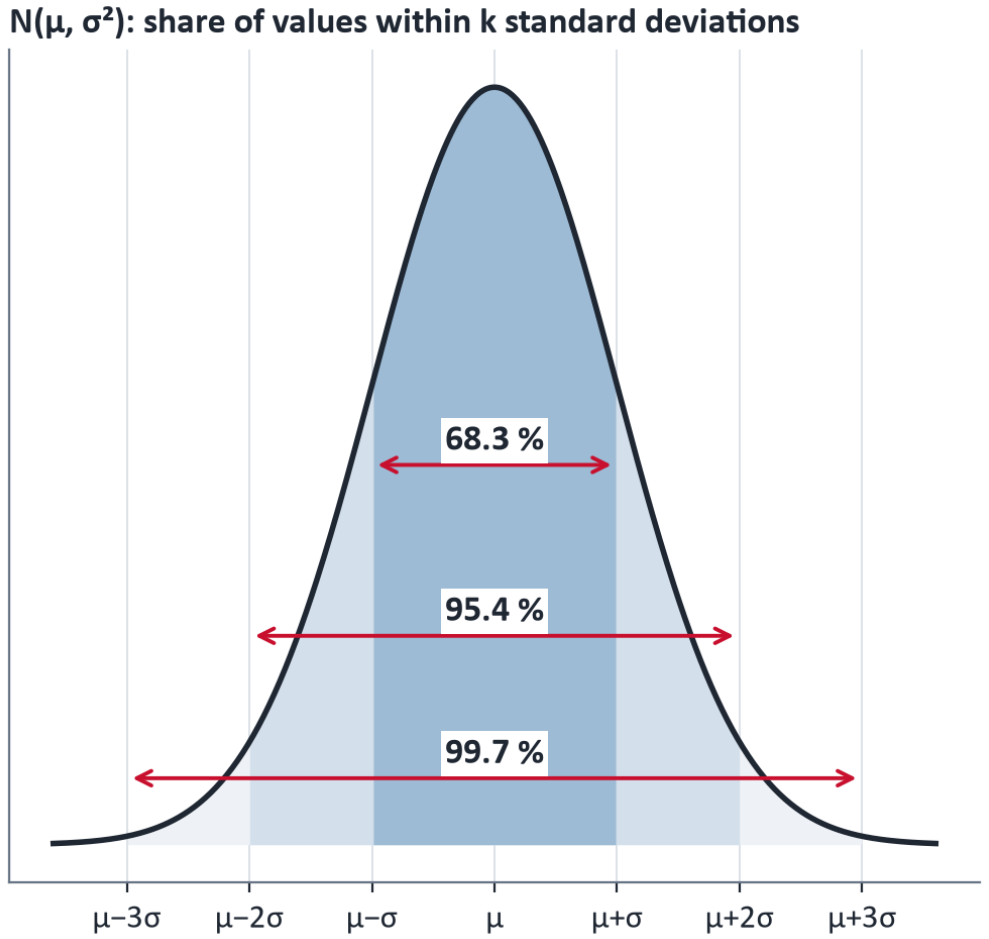


Figure 7 — shares within ±1, 2 and 3 σ

03

Relationships and honest charts

Two variables at a time: how to see and measure a relationship, why totals can mislead, and how to draw charts that do not.



Scatter plots and the correlation coefficient r

- A *scatter plot* draws one point per observation, with one variable on each axis.
- *Pearson's correlation* r measures how close the points lie to a **straight line**; it ranges from −1 to +1.
  - r > 0: both rise together. r < 0: one falls as the other rises. r ≈ 0: no linear trend.
- |r| near 1: a tight line. |r| ≈ 0.3: a weak trend, hard to see by eye.
- Practise: estimate r for each panel before reading its label.

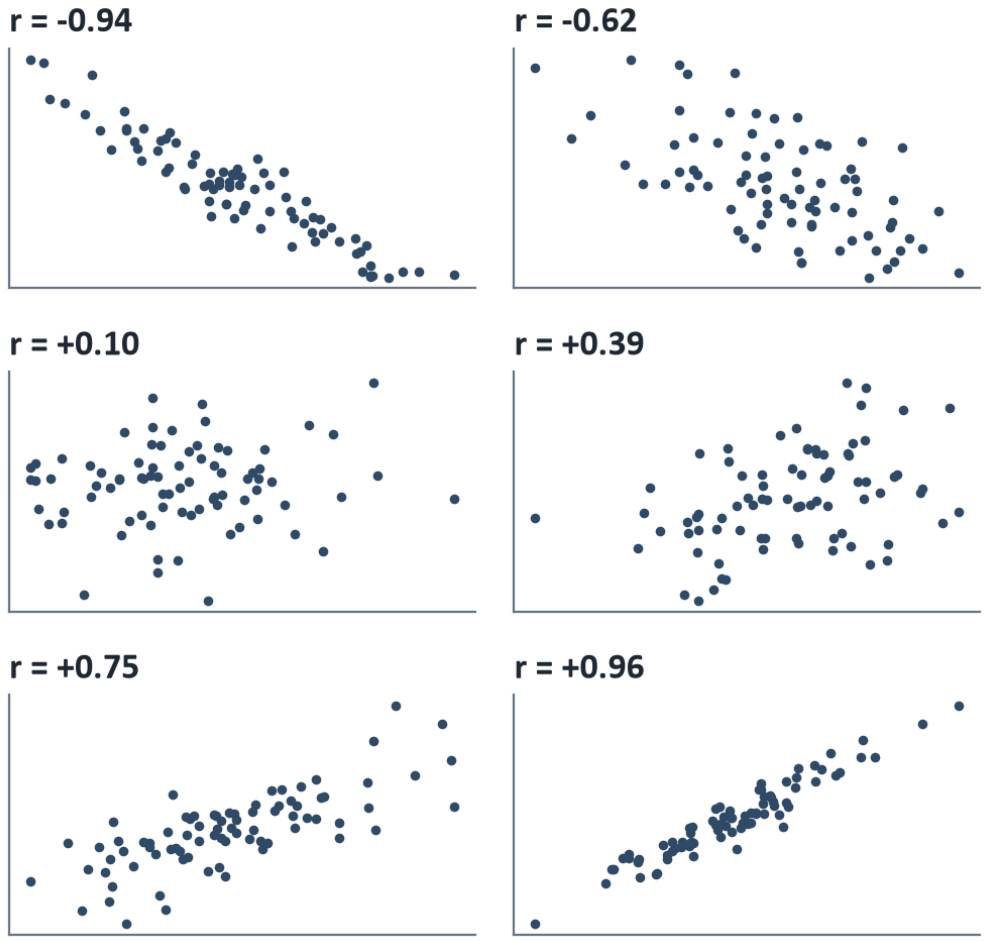


Figure 8 — six scatter plots, n = 80 each

Pearson's r, Spearman's ρ and the limits of correlation

- Pearson's r = s_xy / (s_x · s_y), the *covariance* scaled to [−1, 1]; it captures **linear** relationships only.
- *Spearman's ρ* is r computed on the **ranks**: it captures any relationship that keeps rising (*monotone*) and is less affected by extreme values.
- In the figure, a curved but rising pattern gives r = 0.87 and ρ = 0.94; a parabola gives **r = 0.06** although y depends strongly on x.
- **Correlation is not causation**: ice-cream sales and drownings rise together because both rise in **summer**.

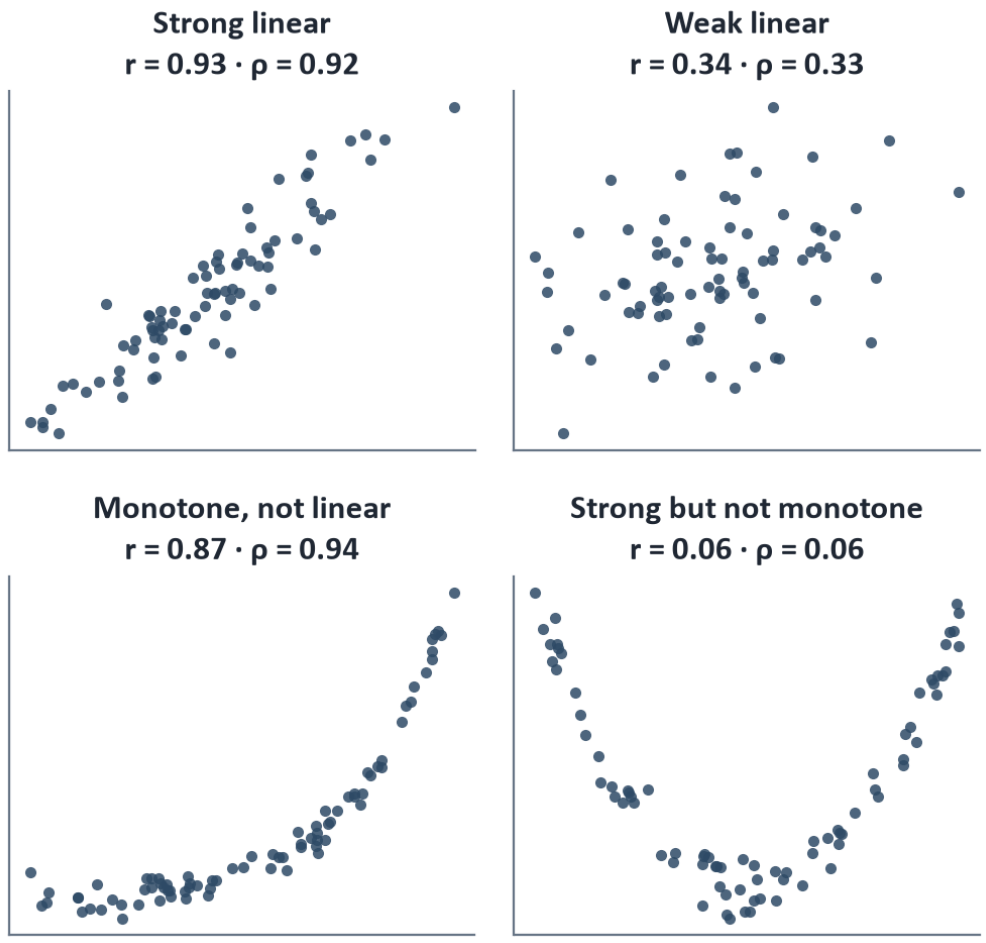


Figure 9 — four relationships, n = 80 each

Simpson's paradox: a comparison that reverses in total

- Two kidney-stone treatments, A and B, 350 patients each (Charig et al., 1986).
- **Small stones**: A succeeds more often, 93 % against 87 %.
- **Large stones**: A again succeeds more often, 73 % against 69 %.
- **All patients**: B looks better, 83 % against 78 %.
- Why: A was given mostly to the **harder large-stone cases** (263 of its 350); the mix of cases, not the treatment, drives the total.
- Lesson: before comparing totals, ask **what else differs between the groups**.

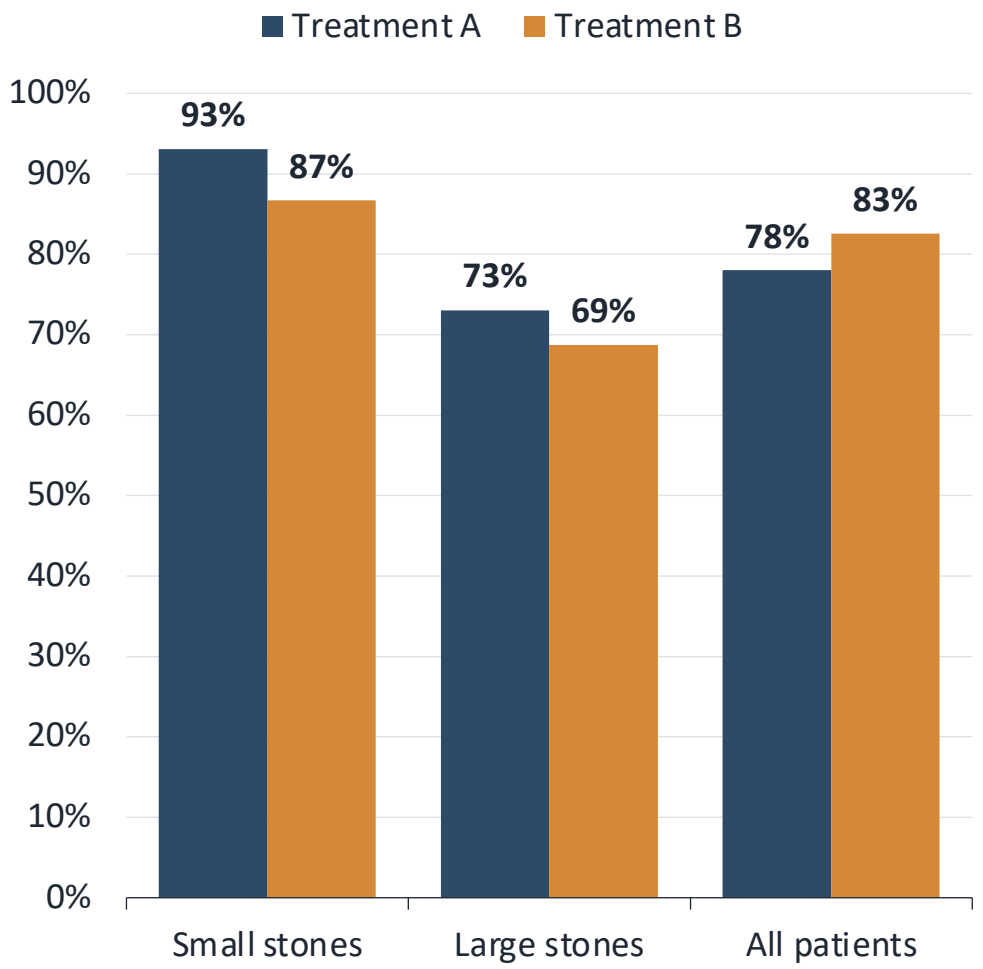


Figure 10 — success rates by stone size

Simpson's paradox in practice: Berkeley admissions, 1973

- Overall, **45 %** of 2,691 men but only **30 %** of 1,835 women were admitted to the six largest departments.
- Department by department, women were admitted at a **higher** rate in 4 of 6.
- The reason: **51 %** of the men but only **7 %** of the women applied to departments A and B, which admit most applicants.
- Splitting by group (*stratification*) reveals this mix effect; whether the comparison is fair is a **causal question**, not only a statistical one.

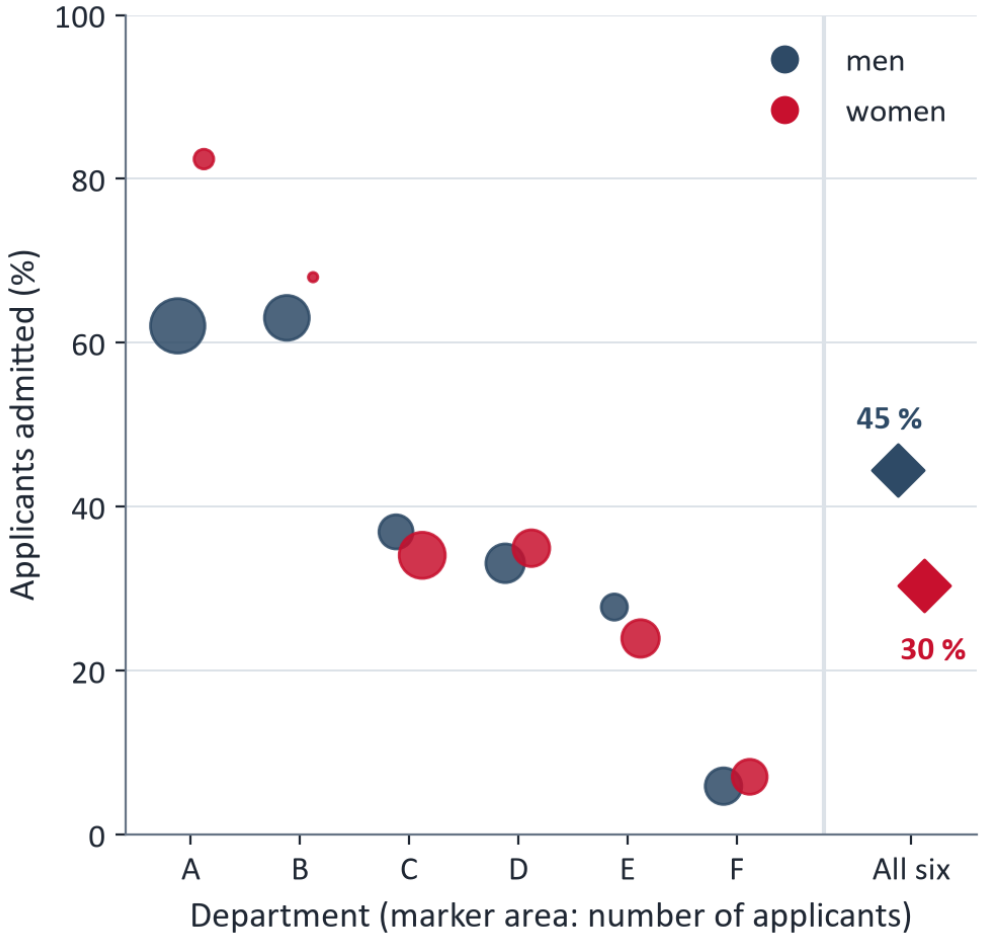


Figure 11 — admission rates, Berkeley 1973

Bar charts versus pie charts for composition

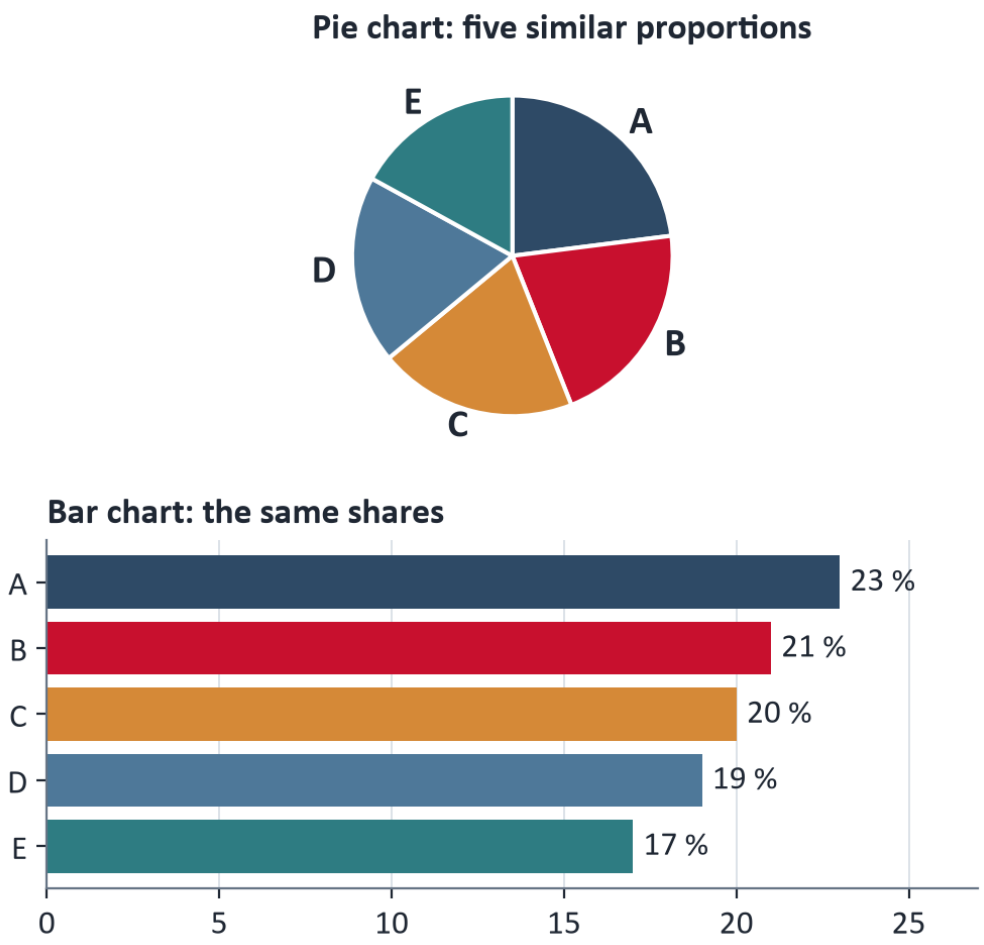


Figure 12 — the same five shares twice

- A pie chart encodes shares as **angles and areas**, which the eye compares poorly.
- A bar chart uses **lengths on a common scale**, which the eye compares accurately (Cleveland and McGill, 1984).
- Try it: the shares 23, 21, 20, 19 and 17 % are hard to rank in the pie and easy in the bars.
- Use a pie only for **two or three parts** of a whole; use a table when exact values matter.

Misleading charts: the truncated axis and its correction

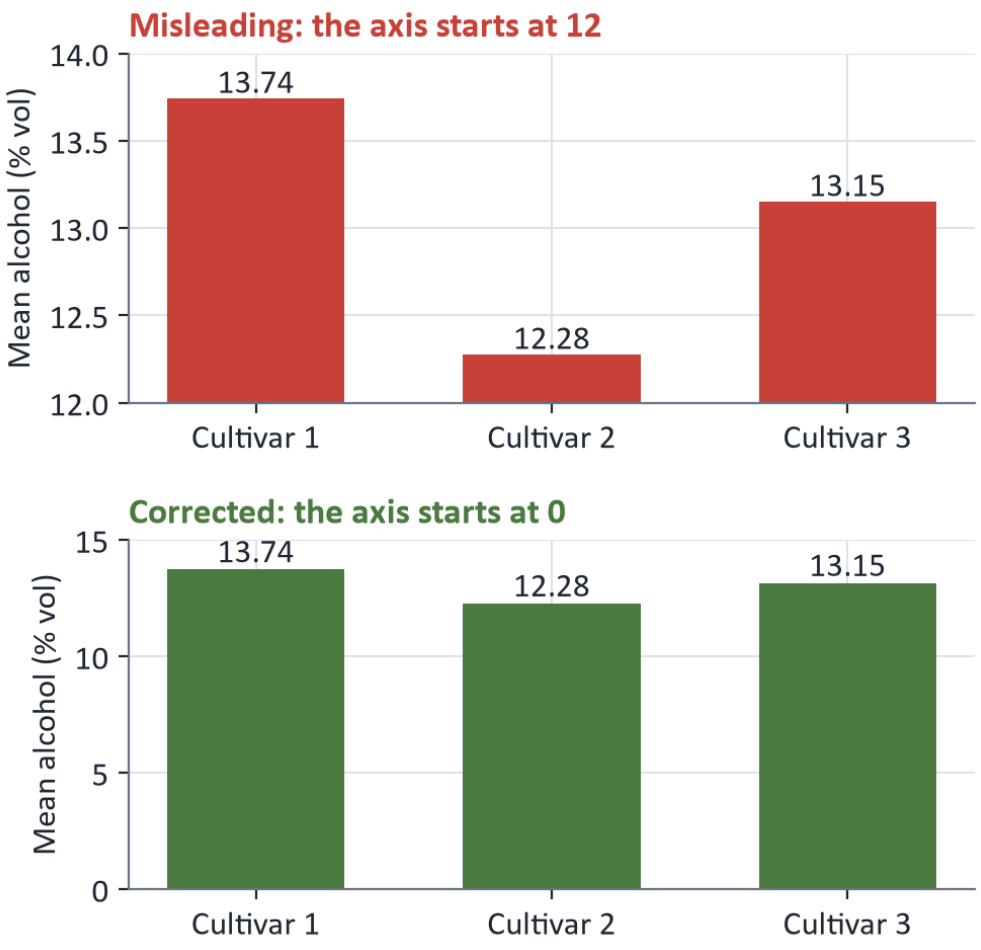


Figure 13 — mean alcohol by cultivar

- A bar shows its value by its **length**, so a bar chart's axis must **start at zero**.
- With the axis starting at 12, cultivar 1 looks **6.3 times** taller than cultivar 2; the true ratio is **1.12**.
- Other tricks: two y-axes with unrelated scales, a hand-picked time window, 3-D effects.
- Dot plots and line charts may start above zero, because **position**, not length, carries the value.

04

From a sample to the population

We see one sample but want to learn about everyone. How much would our numbers change with a different sample?



Population and sample: who is measured matters

- A *population* is everyone we want to know about; a *sample* is the part we actually **observe**.
- *Selection bias*: the way the sample is chosen makes it **systematically different** from the population.
- **1936**: the *Literary Digest* collected 2.4 million answers from telephone and car-owner lists (wealthier voters) and predicted a **defeat for Roosevelt**; he won in a landslide.
  - Gallup, with about **50,000** more representative people, predicted the right winner.
- **A bigger sample does not fix a biased one**: in the figure, the biased sample gives 25.7 against the true 35.3.

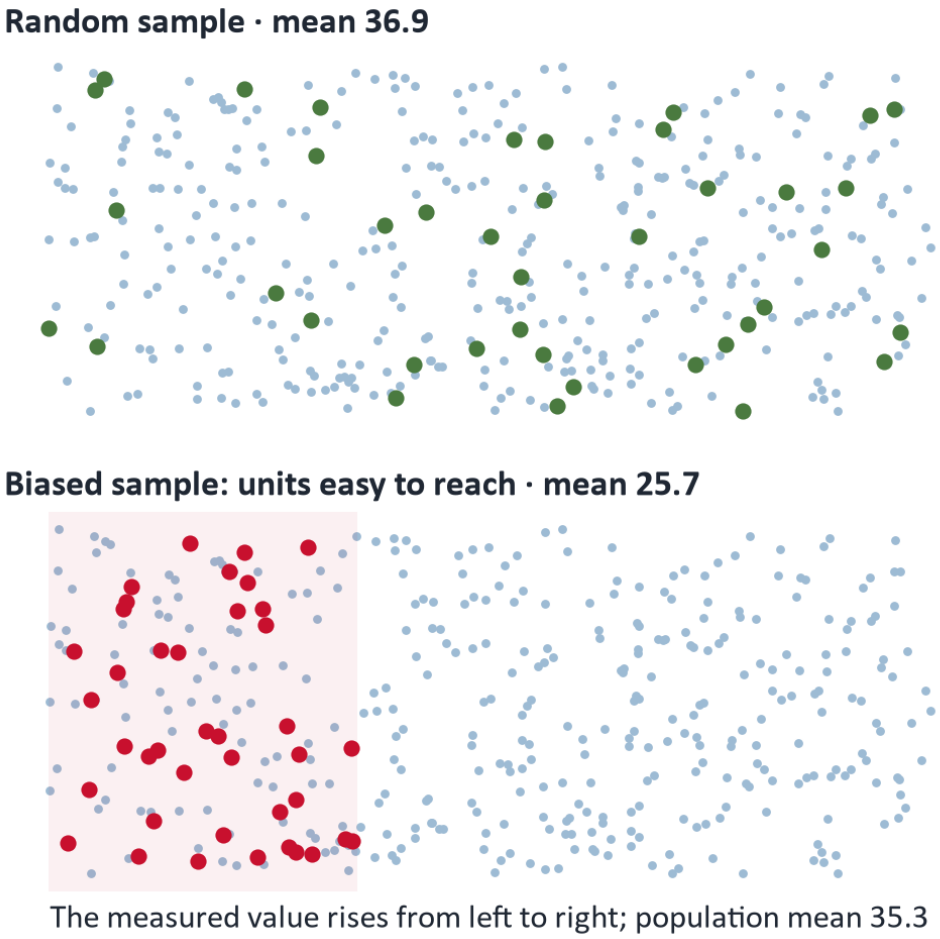


Figure 14 — random and biased samples

The key idea: a statistic changes from sample to sample



- In practice we have **only one sample**. The next slides show two ways to learn the sampling distribution anyway: mathematics (the central limit theorem) and computation (the bootstrap).

The sampling distribution of a statistic

- Take many samples of 25 deliveries and compute the median of each: the medians **differ** (figure, bottom).
- Their histogram is the *sampling distribution* of the median.
- Its standard deviation is the *standard error* (SE): here about **2.3 min**, the typical error of one sample median.
- In practice we have **one sample**, not thousands.
  - Route 1: theory (the central limit theorem, next slide).
  - Route 2: computation (the **bootstrap**).

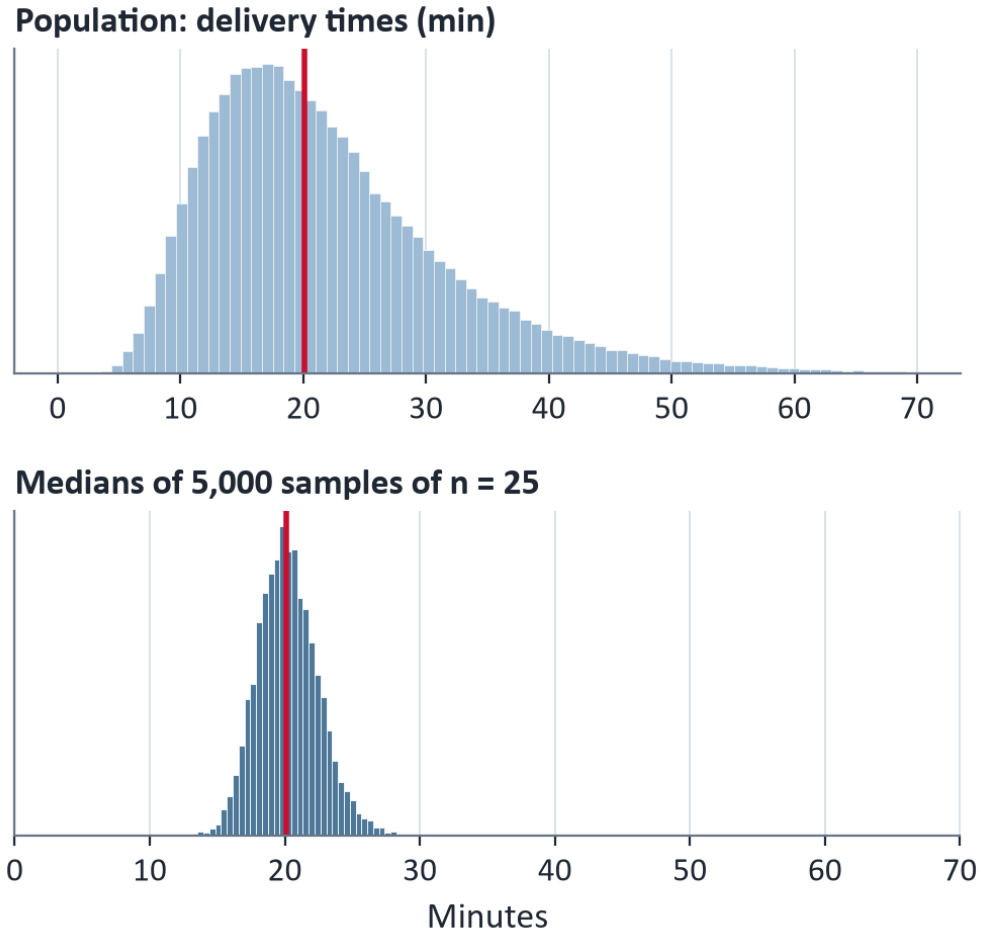


Figure 15 — idealised repeated sampling

The central limit theorem: averages become normal

- In words: the **mean of many independent values** is approximately **normally** distributed, whatever the shape of the data.
- The figure starts from very skewed data (n = 1); means of 2, 5 and 30 values look more and more like the red normal curve.
- The spread of the mean shrinks with n: SE = σ / √n, so **four times more data halves the error**.
- This is why the normal curve appears in tests and intervals for means.
- Caution: for very skewed data, n = 30 may still not be enough.

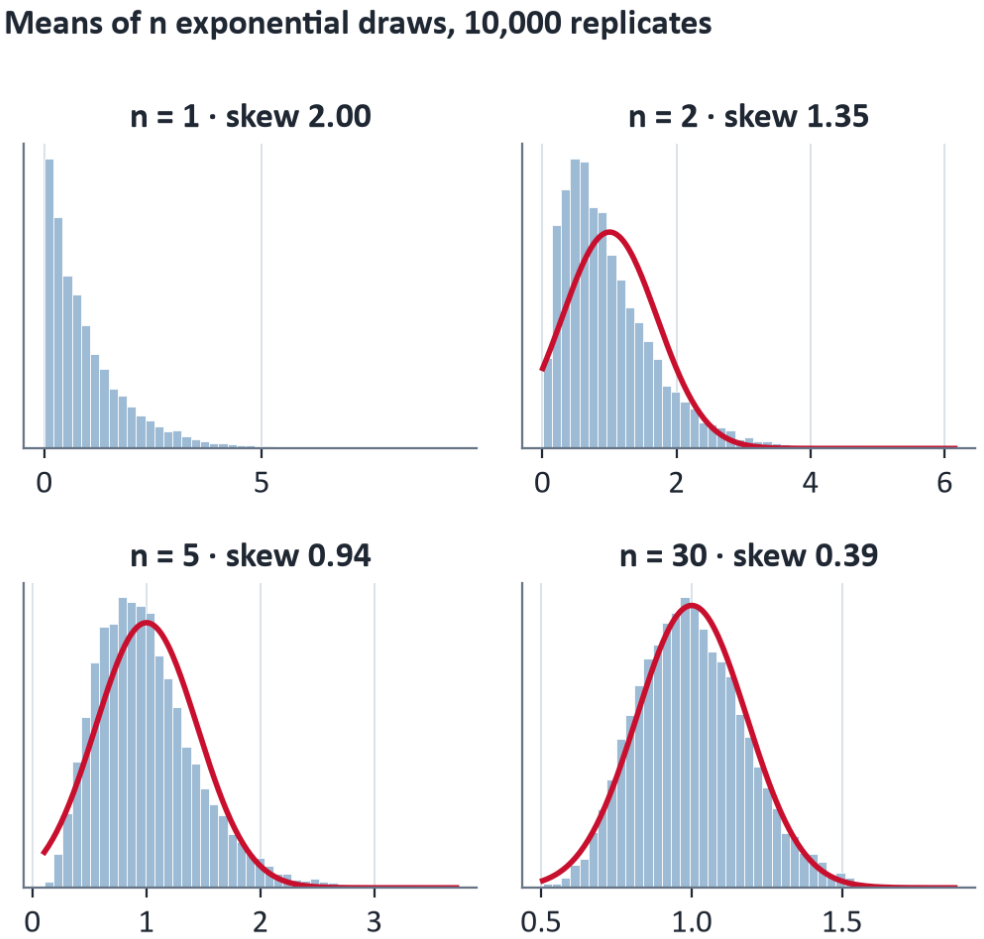


Figure 16 — red: the normal approximation

The bootstrap: resampling the observed sample with replacement

1 Sample

The n observed values stand in for the population.

2 Resample

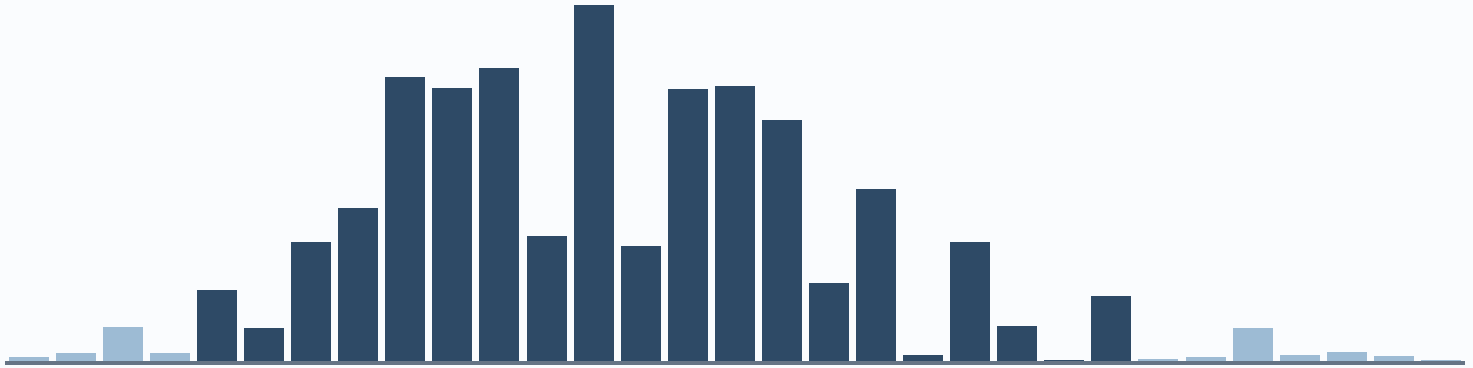
Draw n values with replacement, B = 10,000 times.

3 Recompute

Compute the statistic on every resample.

4 Interval

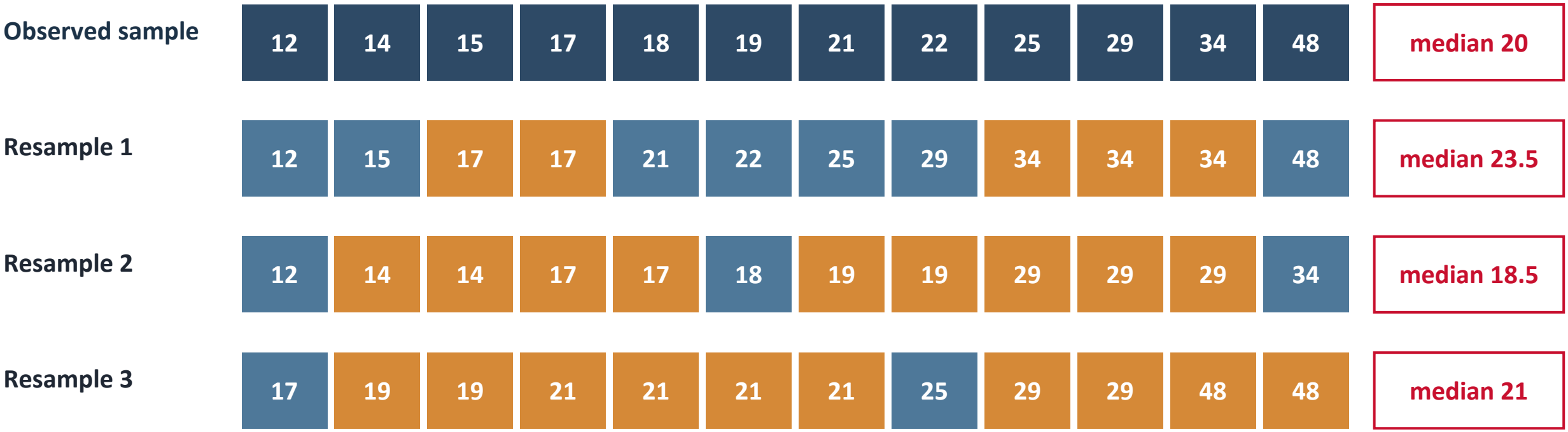
The 2.5 % and 97.5 % quantiles give a percentile interval with nominal 95 % confidence.



10,000 bootstrap medians of the 12 delivery times; dark bars lie inside the 95 % interval [16.0, 27.0]

- The bootstrap (Efron, 1979) treats the sample as a stand-in for the population. Each resample repeats some values and leaves out others (about 35 % on average). It works only if the sample **represents the population well**.

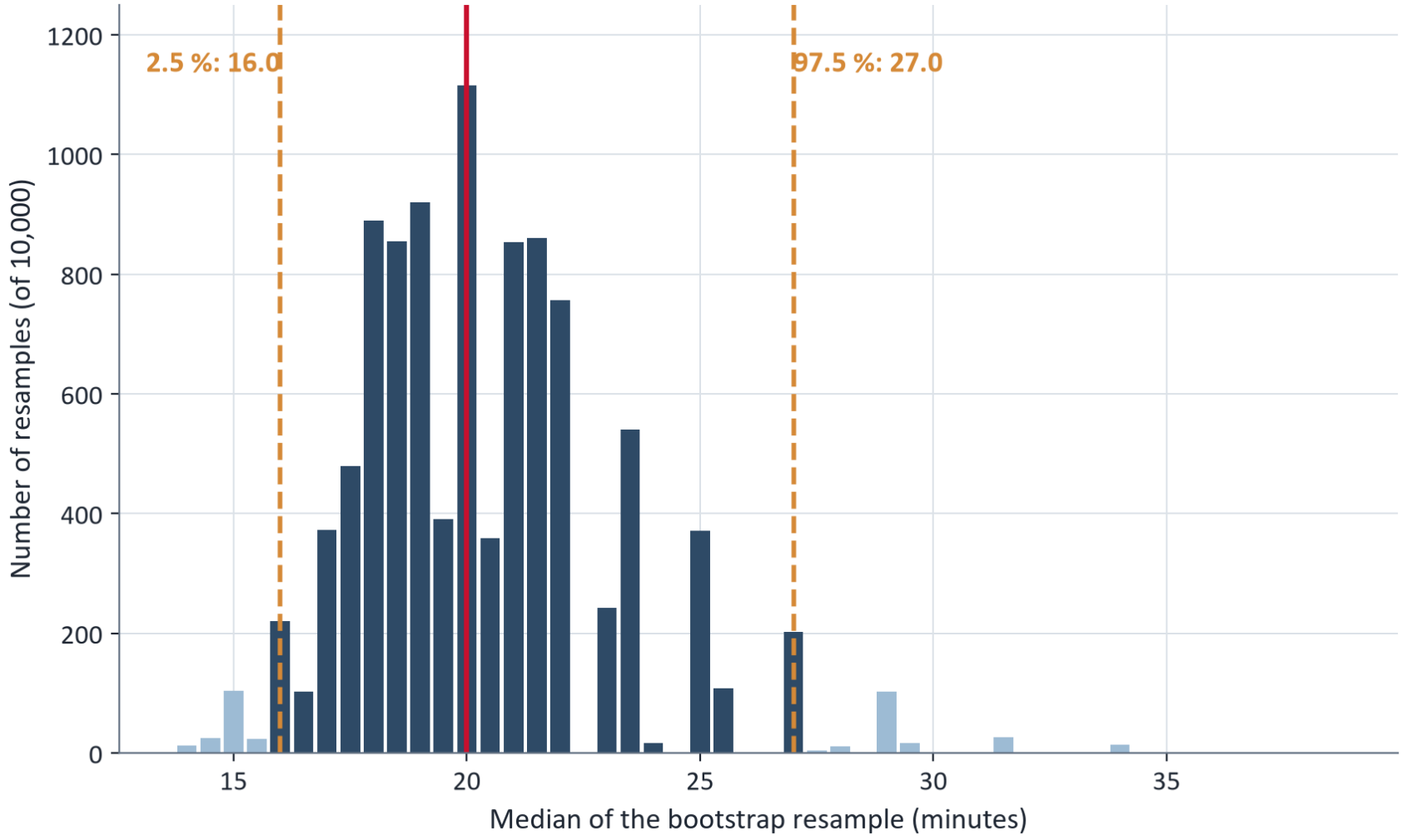
Worked example: three bootstrap resamples of twelve values



Delivery times in minutes, sorted within each row. Gold: a value drawn more than once. These illustrative resamples follow the same rule as the independently generated 10,000 bootstrap replicates.

Figure 17 — each resample draws 12 values with replacement from the observed 12, and the median is recomputed for each

Bootstrap distribution of the median and its 95 % interval



- The sample median is **20.0 min**.
- The 10,000 bootstrap medians show how much the median could vary.
- The middle 95 % gives the interval **[16.0, 27.0] min**.
- More resamples make the histogram smoother but **add no information**: the sample still has 12 values.

What a 95 % confidence interval means

- Instead of one number ("the mean is 50.5"), give a **range**: "between 47.5 and 53.5".
- "95 %" describes the **method**: used on many samples, it captures the true value about **95 times in 100**.
- In the figure, 38 of 40 intervals contain the true mean μ = 50; the 2 red ones miss.
- **One** computed interval either contains μ or not; we cannot know which.

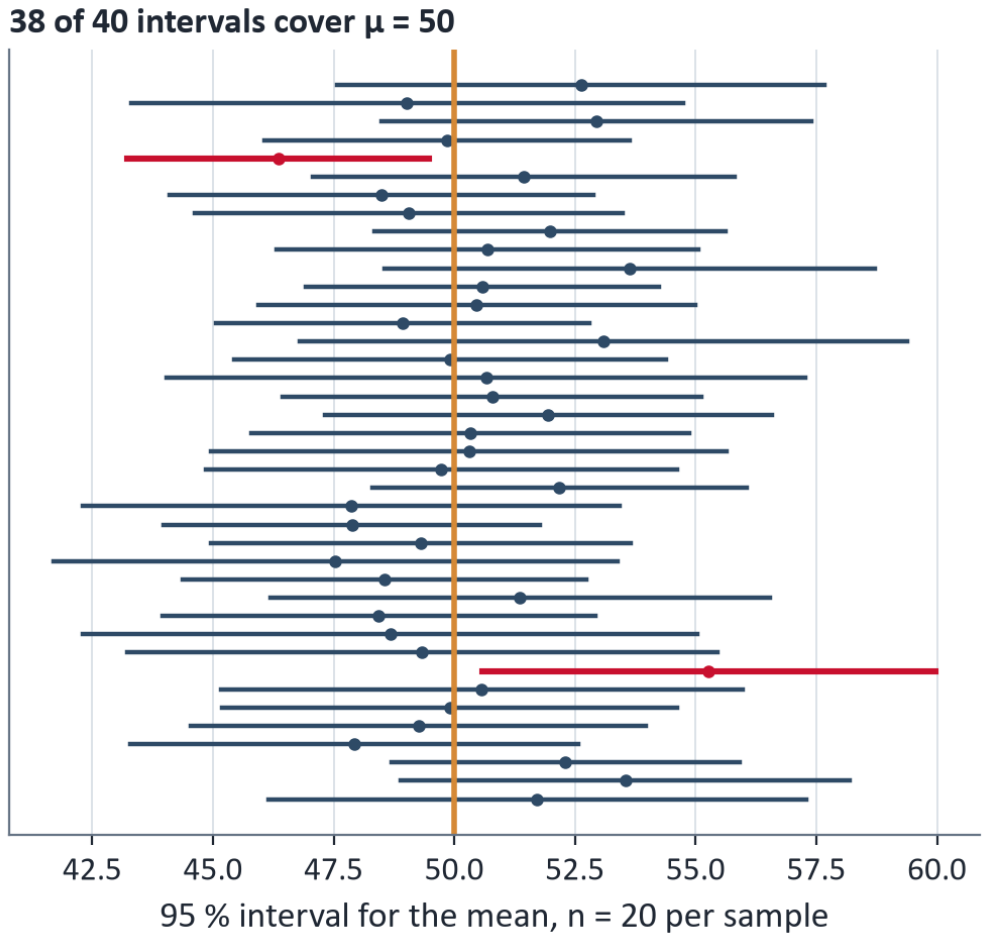


Figure 18 — red: intervals that miss μ

Reading and reporting a confidence interval

1 Report the interval

Write "20 min (95 % CI 16–27)", not just "20 min".
The single number hides how uncertain it is.

2 More confidence → wider

A 99 % interval is wider than a 95 % interval from the same data.
More certainty costs precision.

3 More data → narrower

The width shrinks like 1 / √n.
Four times more data gives an interval about half as wide.

- A wide interval says "be careful: the data fit very different values"; a narrow one says the estimate is precise, but only if the sample is **not biased**.

Two routes to an interval: formula versus bootstrap

- The same 12 delivery times: 12, 14, 15, 17, 18, 19, 21, 22, 25, 29, 34, 48.
- **Formula (t)** for the mean: $\bar{x} \pm t \cdot s / \sqrt{n} = 22.8 \pm 2.20 \times 10.2 / \sqrt{12}$ → **[16.4, 29.3]**; always symmetric, assumes roughly normal data or a large n.
- **Bootstrap** for the mean: **[17.9, 28.9]**, shorter below and longer above, following the **skew**.
- The median has no simple formula, but the bootstrap works the same way: **[16.0, 27.0]**.
- With only 12 values, neither "95 %" is guaranteed.

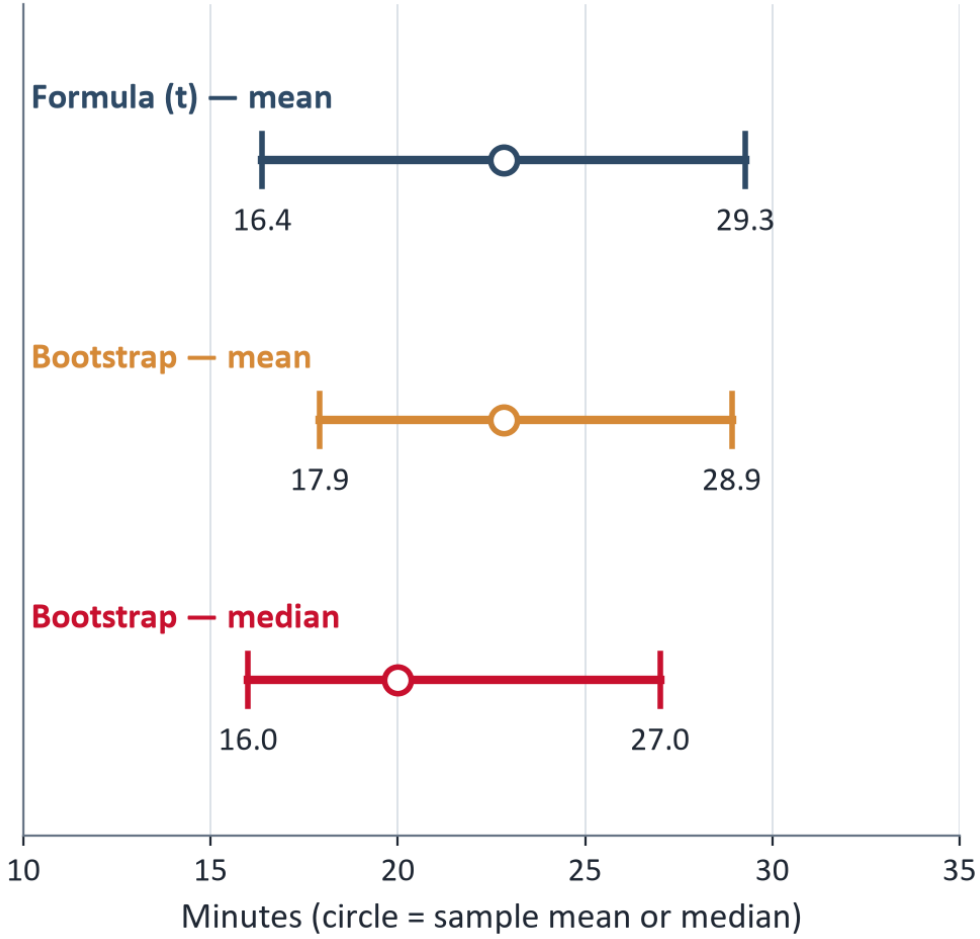


Figure 19 — three 95 % intervals, same 12 values

05

Hypothesis testing

Is the pattern in our sample real, or could chance alone have produced it? A test answers with a clear, repeatable procedure.



The logic of a test: a courtroom analogy

In court	In a hypothesis test (does the new routing app save time?)
The defendant is presumed innocent	*Null hypothesis* H_0: the app has **no effect**
The prosecutor claims guilt	*Alternative* H_1: the app **changes** delivery times
Evidence is presented	Data are collected and a *test statistic* is computed
Guilty only beyond reasonable doubt	Reject H_0 only if the *p-value* is below α (often 0.05)
"Not guilty" does not mean innocent	"H_0 not rejected" does **not** mean H_0 is true
Convicting an innocent person	*Type I error*: a false alarm
Acquitting a guilty person	*Type II error*: a missed effect

- The test never proves H_0. It only asks whether the data are **surprising enough** to abandon it.

Every hypothesis test follows the same five steps

1 Hypotheses

H_0: the app saves no time.
H_1: the app changes the time.

2 Choose α

Fix the risk of a false alarm before seeing the data: $\alpha = 0.05$.

3 Statistic

Measure the signal against the noise:
$t = 2.30$.

4 p-value

If H_0 were true, how often would t be this extreme?
$p = 0.033$.

5 Decide

$p < \alpha \rightarrow$ reject H_0. Report the size of the effect and its interval too.

- Steps 1 and 2 happen **before** looking at the data; changing them afterwards invalidates the test.

The p-value: how surprising are the data if H_0 is true?

- 20 couriers each drive with and without the app; we record the time saved by each courier.
- Test statistic t = (mean saving − 0) / (s / √n) = **2.30**: the saving is 2.3 standard errors away from zero.
- *p-value*: if the app had **no effect**, the chance of a t at least this far from zero is **0.033** (red tails).
- p = 0.033 < α = 0.05 → **reject H_0**: chance alone is a poor explanation.
- The p-value is **not** the probability that H_0 is true, and a large p does **not** prove there is no effect.

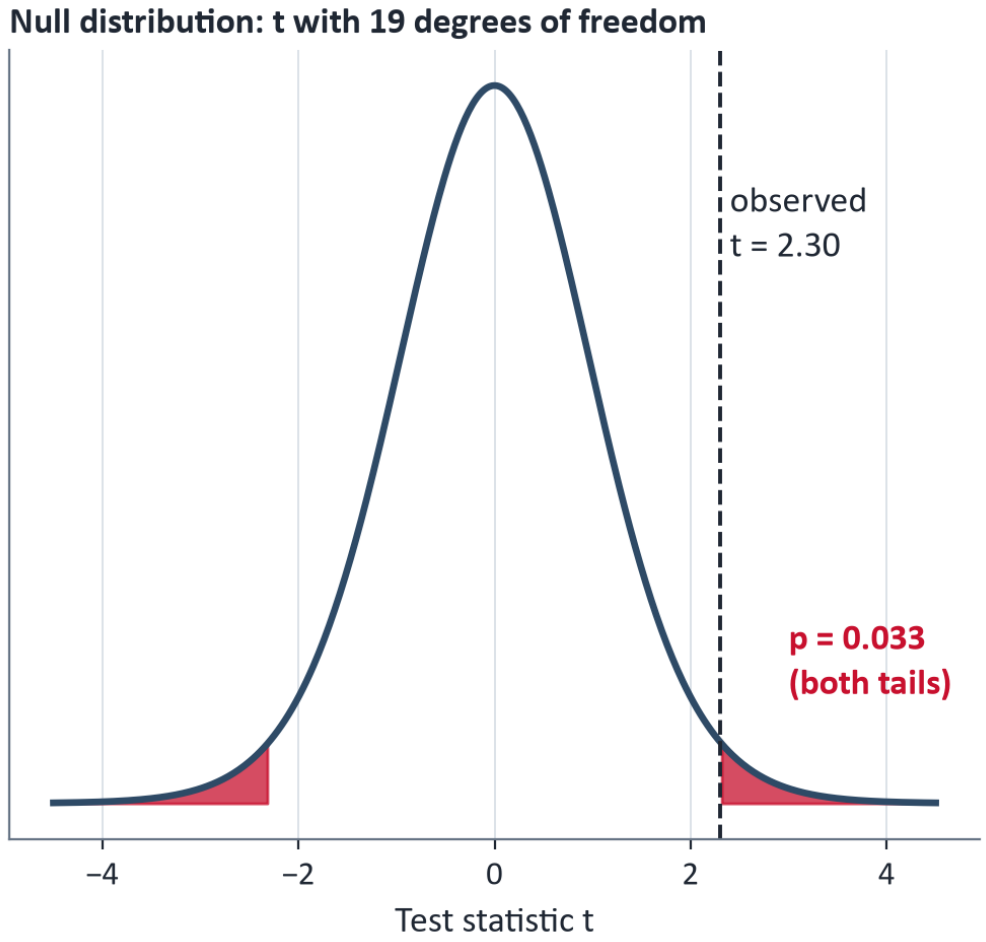


Figure 20 — the p-value as a tail area

Two ways a test can be wrong

	Reality: H_0 true (no effect)	Reality: H_0 false (real effect)
Decision: reject H_0	**Type I error: false alarm** probability α a fire alarm without a fire	**Correct: effect detected** probability $1 - \beta$ = power
Decision: do not reject H_0	**Correct: no false alarm** probability $1 - \alpha$	**Type II error: missed effect** probability β a sick patient sent home

- We control α directly by choosing it. β depends on the **sample size** and on **how large the real effect is**.

The critical value, α, β and the power of a test

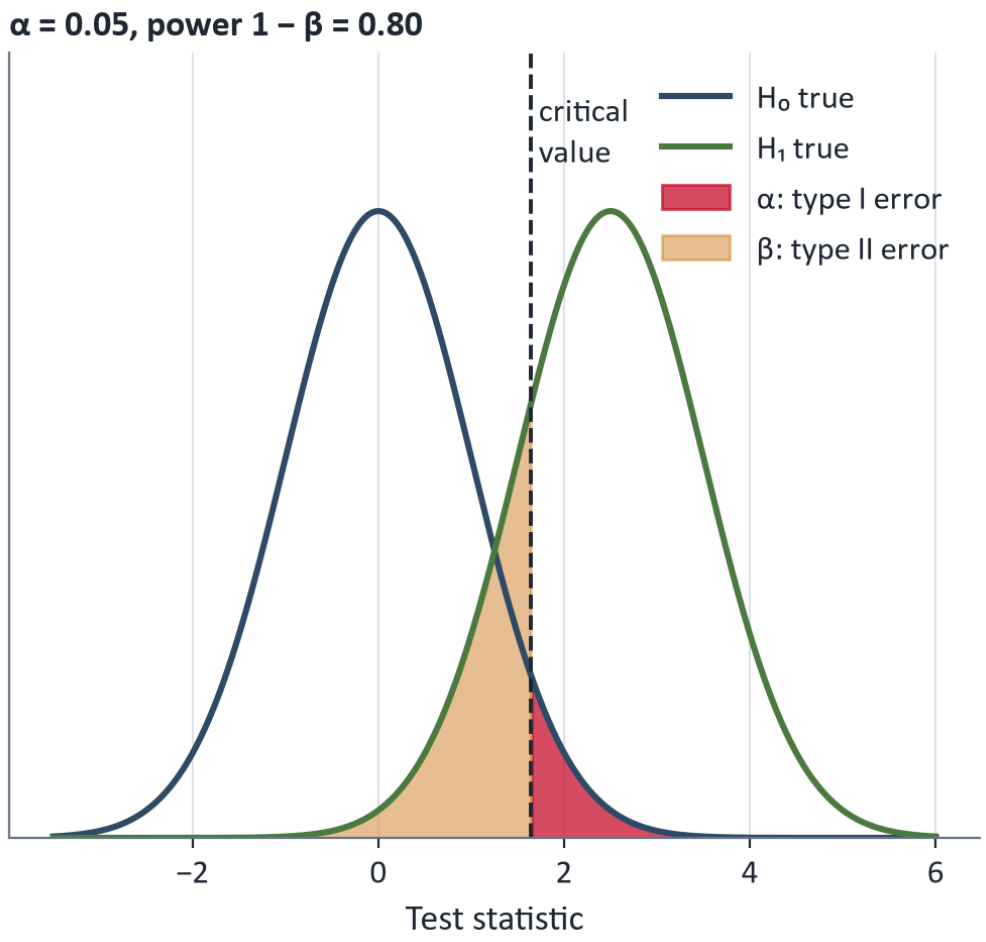


Figure 21 — α, β and the critical value

- The *critical value* is the cut-off for the test statistic: for α = 0.05 (one-sided) under the normal curve it is **1.65**.
  - Statistic above 1.65 → reject H_0; below → "not enough evidence".
- Red area = **α**: a false alarm when H_0 is true.
- Gold area = **β**: missing a real effect; the *power* 1 − β is **0.80** here.
- Power grows with **larger samples** and **larger effects**; a stricter α makes β larger.

Worked example: the two-sample t-test

- The *two-sample t-test* asks whether two groups have **different means**.
- Example: task completion times of 25 users per design; means **52.0 s** and **45.3 s**.
- t = difference / its standard error = 6.7 / 2.29 = **2.93**: the difference is about **3 times** the wobble expected from chance.
- p = 0.005 < 0.05 → reject H_0; the difference is 6.7 s, 95 % CI **[2.1, 11.3] s**.
- Welch's version (used here) does not assume equal spreads; it needs **independent** observations.

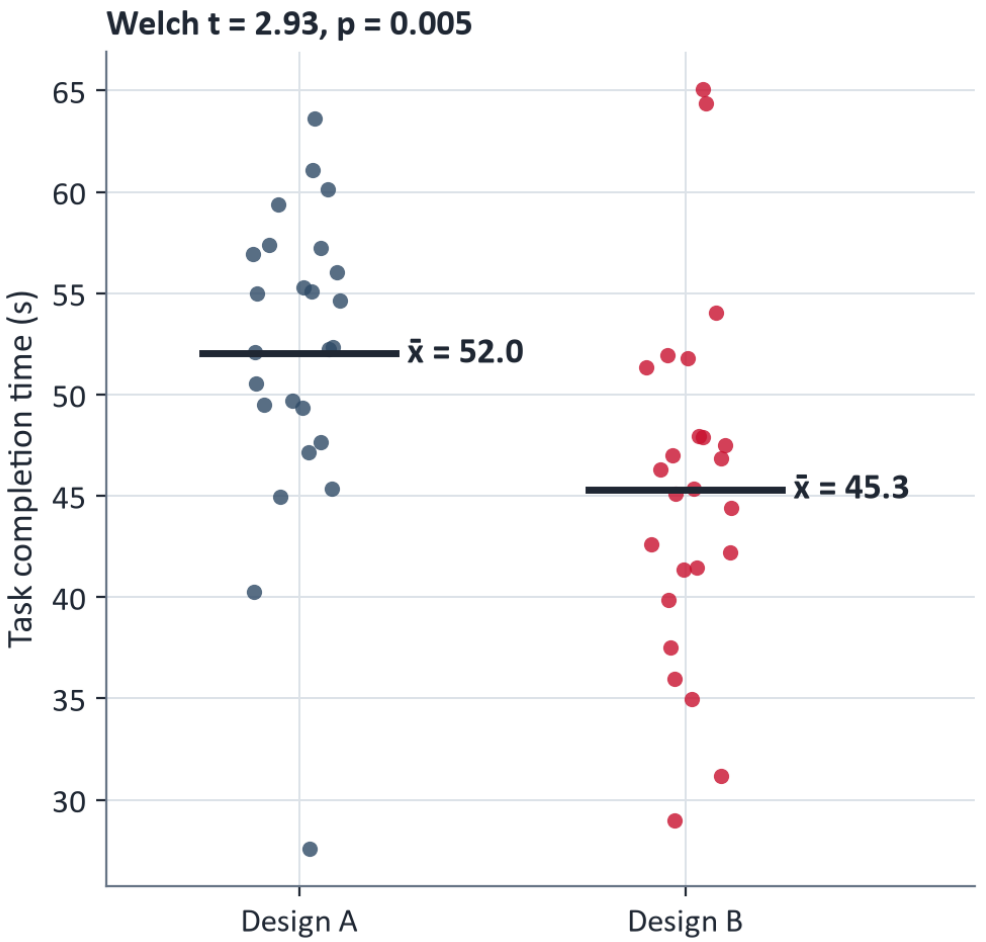


Figure 22 — simulated completion times

Worked example: the chi-squared test for counts

- Claim: "**70 %** of our deliveries arrive on time." We check 60 deliveries.
- Under H_0 we expect **42** on time and **18** late; we observe **50** and **10**.
- $\chi^2 = \Sigma\ (\text{observed} - \text{expected})^2 / \text{expected} = 8^2/18 + 8^2/42 =$ **5.08**; with 1 degree of freedom, p = **0.024**.
- $p < 0.05 \rightarrow$ the 70 % claim does not fit: deliveries are on time **more** often (83 %).
- Needs independent observations and at least about **5 expected** per category.

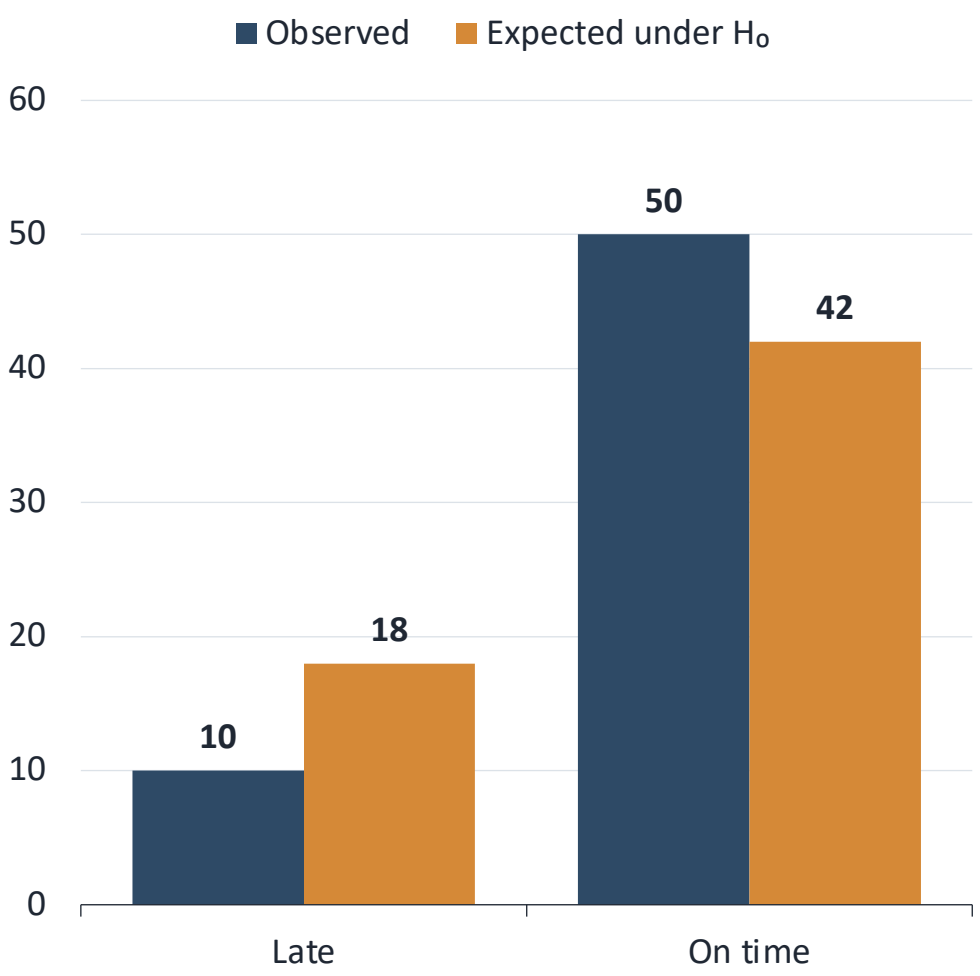


Figure 23 — observed and expected counts

06

AI-assisted exploratory analysis

Automated profiling and notebook agents support exploratory workflows; analysts remain responsible for data provenance, statistical validity and verification.



Automated profiling: per-column summaries and alert rules

UCI Wine data: 178 rows, 13 columns, 0 missing values, 0 duplicated rows

Column	Missing	Distinct	Mean	SD	Median	Skewness	Alert
alcohol	0	126	13.00	0.81	13.05	-0.05	—
malic acid	0	133	2.34	1.12	1.87	+1.03	**skewed**
magnesium	0	53	99.74	14.28	98.00	+1.09	**skewed**
total phenols	0	97	2.30	0.63	2.35	+0.09	**r = 0.86 flavanoids**
flavanoids	0	132	2.03	1.00	2.13	+0.03	**r = 0.86 total phenols**
proline	0	121	746.89	314.91	673.50	+0.76	—

Alert rules of this example

Skewed
|skewness| > 1

Correlated
|r| > 0.8 with another column

Missing
fraction of missing values > 0.10

Constant
a single distinct value

Duplicates
identical rows present

- A profile report summarises every column, the correlations, missing values and duplicates in one step. The alert thresholds are rules of thumb: a profile shows **what to check**, not what to conclude.

A review checklist for an AI-generated exploratory analysis

1 Data

The right file, all rows loaded, types parsed correctly.

2 Preparation

Missing values, filters, joins, duplicates and sorting stated and justified.

3 Statistics

Each statistic suits the variable type and the shape of its distribution.

4 Charts

Bars start at zero; bins, axes and units are stated.

5 Claims

Every sentence is traced to an output; the number of tests run is reported.

6 Repetition

Several independent runs; the spread of the estimates is reported.

- The checklist follows the order of the analysis; most disagreements between AI analyses arise in data preparation (Cui and Alexander, 2026). Passing it makes an analysis auditable, not automatically correct: a biased sample stays biased.

Key takeaways

- ✓ Know the **variable type** first, and plot the distribution before computing.
- ✓ The mean and SD suit symmetric data; the **median and IQR** resist skew and outliers.
- ✓ Correlation measures a **linear** link, not causation; groups can reverse a trend (Simpson).
- ✓ A statistic varies from sample to sample; the **CLT and the bootstrap** measure that variation.
- ✓ Report a **confidence interval**, not just a number: "20 min (95 % CI 16–27)".
- ✓ A p-value: how surprising the data are **if H_0 were true**, not the probability that H_0 is true.
- ✓ An AI-generated analysis is a **draft** to review with a checklist.



Questions

Co-funded by the European Union through the EuroHPC Joint Undertaking under Grant Agreement No 101263128, and by the Ministry of Digital Transformation of the Republic of North Macedonia. Views and opinions expressed are those of the author(s) only and do not necessarily reflect those of the European Union or the EuroHPC JU.

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EuroHPC Joint Undertaking

Co-funded by the European Union

Republic of North Macedonia
Ministry of Digital Transformation